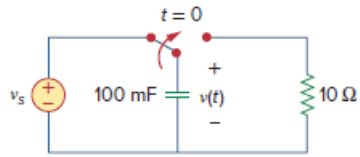


## Problem

Find  $v(t)$ ,  $t > 0$  in the circuit of Fig. 16.41. Let  $v_s = 12$  V.

**Figure 16.41:**



## Step-by-step solution

## Step 1 of 8

Refer to Figure 16.41 in the text book.

For  $t < 0$ , the capacitor acts as open circuit in Figure 16.41.

Draw the circuit diagram for  $t < 0$ .

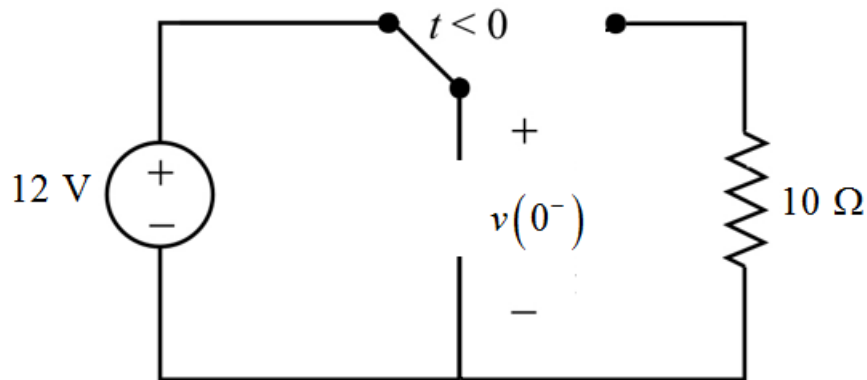


Figure 1

## Step 2 of 8

From Figure 1, the initial voltage across the capacitor is,

$$v(0^-) = 12 \text{ V}$$

The capacitor does not allow sudden changes in the voltage across it. Therefore,

$$\begin{aligned} v(0^-) &= v(0^+) \\ &= v(0) \\ &= 12 \text{ V} \end{aligned}$$

## Step 3 of 8

Draw the circuit diagram for  $t > 0$ .

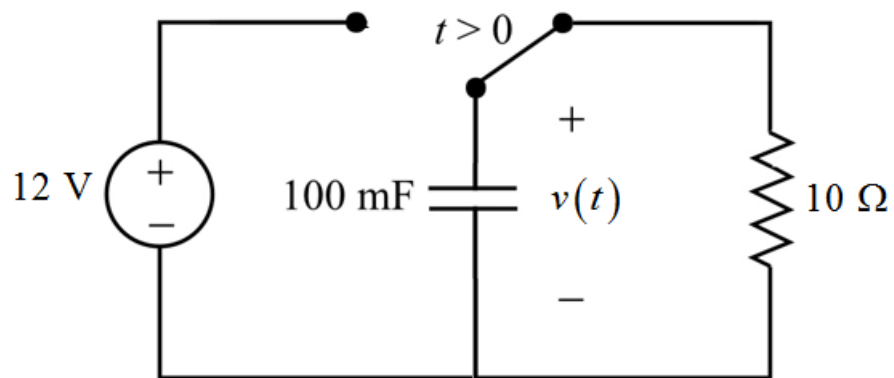


Figure 2

Step 4 of 8

Represent the circuit in s-domain including the initial values.

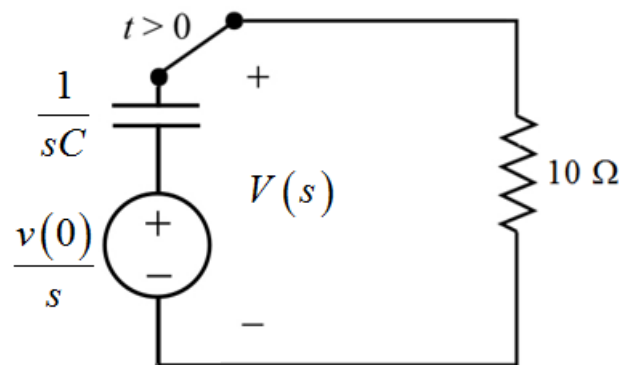


Figure 3

Step 5 of 8

The value of  $\frac{1}{Cs}$  is,

$$\begin{aligned}\frac{1}{Cs} &= \frac{1}{(100 \times 10^{-3})s} \\ &= \frac{10}{s}\end{aligned}$$

The value of  $\frac{v(0)}{s}$  is,

$$\frac{v(0)}{s} = \frac{12}{s}$$

Step 6 of 8

Draw the modified circuit diagram.

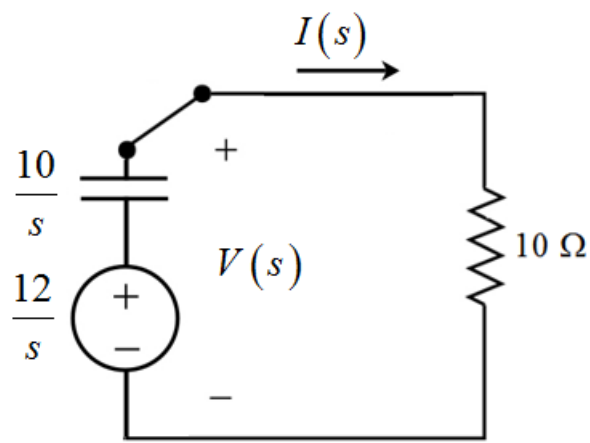


Figure 4

**Step 7** of 8

Calculate the current flowing through the circuit in Figure 4.

$$\begin{aligned} I(s) &= \frac{\frac{12}{s}}{\frac{10}{s} + 10} \\ &= \frac{\frac{12}{s}}{\frac{10s + 10}{s}} \\ &= \frac{12}{10s + 10} \\ &= \frac{1.2}{s + 1} \end{aligned}$$

**Step 8** of 8

Determine the voltage across the capacitor.

$$V(s) = \frac{10}{s}[-I(s)] + \frac{12}{s}$$

Substitute  $\frac{1.2}{s+1}$  for  $I$  in the equation.

$$\begin{aligned} V(s) &= \frac{10}{s} \left[ -\left( \frac{1.2}{s+1} \right) \right] + \frac{12}{s} \\ &= \frac{-12}{s(s+1)} + \frac{12}{s} \\ &= \frac{-12+12(s+1)}{s(s+1)} \\ &= \frac{-12+12s+12}{s(s+1)} \\ &= \frac{12s}{s(s+1)} \end{aligned}$$

$$V(s) = \frac{12}{s+1}$$

Apply the inverse Laplace transform on both sides.

$$v(t) = 12e^{-t}u(t) \text{ V}$$

Thus, the voltage across the capacitor,  $v(t)$  for  $t > 0$  is  $\boxed{12e^{-t}u(t) \text{ V}}$ .